
Modelling Moist Precipitating Convection

— Tyler Do —

Modelling Goals

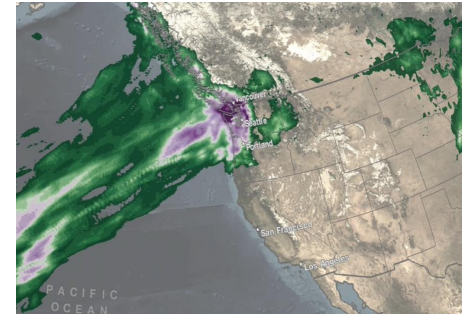
Scattered Convection



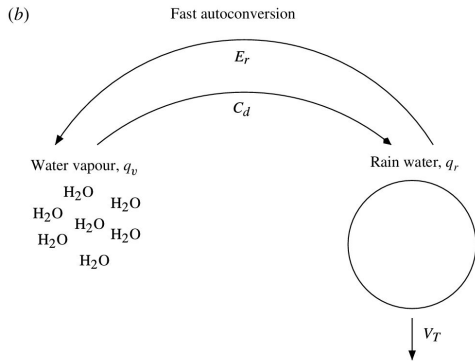
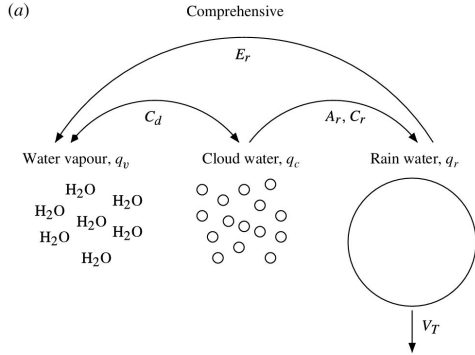
Squall Line



Atmospheric River



Moist Boussinesq Approximation (FARE)



$$\frac{Du}{Dt} = -\nabla\phi + k\,b(\theta_r, q_t, z)$$

$$\nabla \cdot \boldsymbol{u} = 0$$

$$\frac{D\theta_r}{Dt} + \frac{L}{c_p} V_T \frac{\partial q_r}{\partial z} = 0$$

$$\frac{Dq_t}{Dt} - V_T \frac{\partial q_r}{\partial z} = 0$$

$$\theta_e = \theta + \frac{L}{c_p} q_v$$

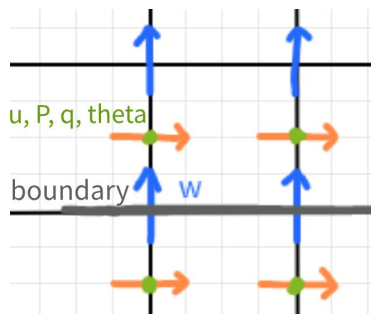
$$\theta_r = \theta - \frac{L}{c_p} q_r$$

$$q_t = q_v + q_r,$$

$$b = g \begin{cases} \frac{\theta_r - \tilde{\theta}(z)}{\theta_o} + \varepsilon_o(q_t - \tilde{q}_v(z)) & \text{if } q_t < q_{vs} \\ \frac{\theta_r - \tilde{\theta}(z)}{\theta_o} + \left(\frac{L}{c_p \theta_o} - 1 \right) (q_t - q_{vs}(z)) + \varepsilon_o(q_{vs}(z) - \tilde{q}_v(z)) & \text{if } q_t \geq q_{vs}. \end{cases} \quad (2.28)$$

Hernandez-Duenas, Gerardo, Andrew J. Majda, Leslie M. Smith, and Samuel N. Stechmann. "Minimal Models for Precipitating Turbulent Convection." *Journal of Fluid Mechanics* 717 (February 2013): 576–611. <https://doi.org/10.1017/jfm.2012.597>.

Spatial Implementation



Pseudo-spectral Fourier Galerkin
(Horizontal, 3/2 rule)

$$\partial_x \hat{u}_k = ik \hat{u}_k \quad \partial_x^2 \hat{u}_{k,j} = -k^2 \hat{u}_{k,j}$$

2nd order Centered Differences
(Vertical, Staggered)

$$\partial_z \hat{u}_{k,j} \approx \frac{\hat{u}_{k,j+1} - \hat{u}_{k,j-1}}{2\Delta z} = \frac{\hat{u}_{k,j+\frac{1}{2}} - \hat{u}_{k,j-\frac{1}{2}}}{\Delta z}$$

$$\partial_z^2 \hat{u}_{k,j} \approx \frac{\hat{u}_{k,j+1} - 2\hat{u}_{k,j} + \hat{u}_{k,j-1}}{\Delta z}$$

Nonlinear Advection

$$u \cdot \nabla \theta = \nabla \cdot (u\theta)$$

(flux form for θ_r, q_t)

$$u \cdot \nabla v = \frac{1}{2}(\nabla \cdot (uv) + u \cdot \nabla v)$$

(skew-symmetric form for u, w)

Pressure Poisson Equation

$$\nabla^2 \hat{P}^* \approx (-k^2 + \partial_z^2) \hat{P}^*_{k,j} = (-k^2 I + D_{zz}) \hat{P}^*$$

(becomes tridiagonal and is no longer the main computational bottleneck)

Time Implementation

Explicit AB3 (Advect, Sources, Forcing)

$$\frac{u^* - u^n}{\Delta t} = \frac{23}{12}f(u^n) - \frac{16}{12}f(u^{n-1}) + \frac{5}{12}f(u^{n-2})$$

Explicit EF (Pressure Forcing)

$$\frac{u^* - u^n}{\Delta t} = f(u^n)$$

Implicit CN (Vertical diffusion)

$$\frac{u^{n+1} - u^*}{\Delta t} = \frac{1}{2}(f(u^*) + f(u^{n+1}))$$

Exact (Horizontal hyperdiffusion)

$$\hat{u}_k^{n+1} = \exp(-\gamma k^4 \Delta t) \hat{u}_k^*$$

Semi-Implicit Advection-Diffusion (θ_r, q_t)

1. Apply all explicit terms, excluding diffusion
 $\theta^n \rightarrow \theta^* \quad q^n \rightarrow q^*$
2. Apply implicit diffusion and exact hyperdiffusion
 $\theta^* \rightarrow \theta^{n+1} \quad q^* \rightarrow q^{n+1}$

Semi-Implicit Incremental Projection (u, w, P)

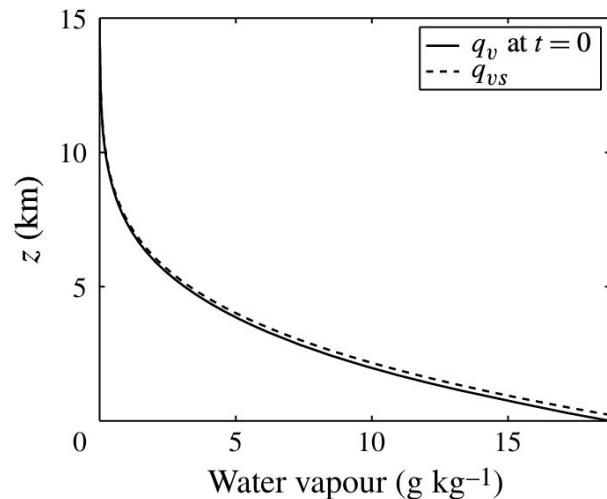
1. Apply all explicit terms, including explicit half of diffusion
 $u^n \rightarrow u^* \quad w^n \rightarrow w^*$
2. Solve incremental PPE for p^*
3. Update $p^n + p^* \rightarrow p^{n+1}$ with p^{n+1} BC
4. Apply p^{n+1} with implicit diffusion and exact hyperdiffusion to u, w
 $u^* \rightarrow u^{n+1} \quad w^* \rightarrow w^{n+1}$

Initial/Boundary/Forcing

- Initial condition: Temperature perturbations near surface with $u_{bg} = 0$

$$\theta_r^0(z) = \tilde{\theta}(z) + \begin{cases} 0 & \text{if } z > 2 \text{ km} \\ X \sim U(-0.1, 0.1) & \text{if } z \leq 2 \text{ km} \end{cases}$$

$$q_t^0(z) = q_v^0(z) = 0.9q_{vs}(z), \quad p^0(z) = \tilde{p}(z), \quad w^0(z) = 0, \quad q_r^0(z) = 0$$



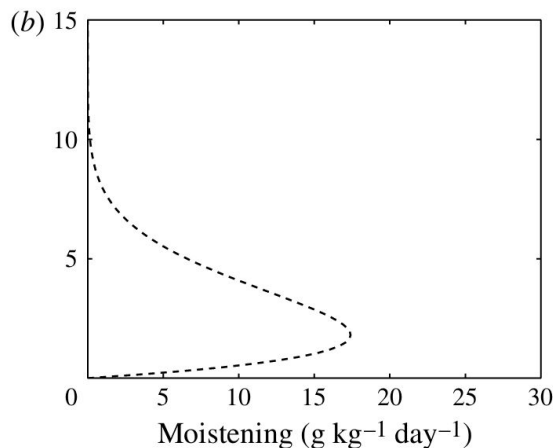
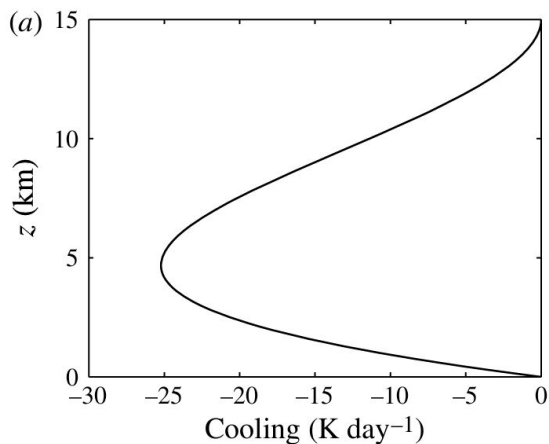
- Boundary conditions: Periodic in x for all variables, zero Dirichlet at both z bounds for w , zero Dirichlet at lower z bound for u , zero Neumann at upper z bound for u , Neumann at both z bounds for all other variables (θ_r, q_v, q_r, q_t) given by:

$$\frac{\partial g}{\partial z} = \frac{\partial \tilde{g}(z)}{\partial z}$$

- Additional forcing and relaxation terms are added to zonal u , θ_r , and q_t equations

Forcing and Relaxation

Forcing Terms



Relaxation Term

$$-\frac{1}{\tau_u}(\bar{u}(z) - u_{bg}(z))$$

Hernandez-Duenas, Gerardo, Andrew J. Majda, Leslie M. Smith, and Samuel N. Stechmann. "Minimal Models for Precipitating Turbulent Convection." *Journal of Fluid Mechanics* 717 (February 2013): 576–611. <https://doi.org/10.1017/jfm.2012.597>.

Simulation Parameters

Physical parameters (all SI or dimensionless units):

$$g = 9.8, \quad \theta_o = 300, \quad B = 0.003, \quad c_p = 1000, \quad H_0 = 12 \text{ km}, \quad \tau_u = 14400, \quad a = 11.11, \quad L = 2.5 \times 10^6, \\ R_d = 287, \quad R_v = 462, \quad p_o = 10^5, \quad \rho_o = \frac{p_o}{R\theta_o} = 1.16, \quad \varepsilon_o = 0.6, \quad q_{vs,o} = 0.02, \quad V_T = 5.5$$

Computational parameters:

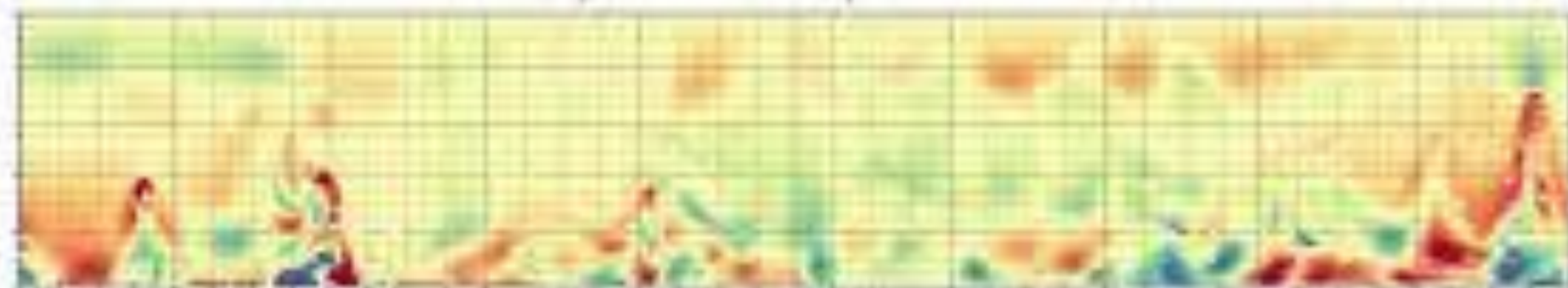
$$\Delta x, \Delta z = 1 \text{ km}, 150 \text{ m} \text{ (256x100 grid)}, \quad \Delta t = 2.2 \text{ s}, \quad T \approx 18000 \text{ s}$$

$$\nu_z = 25 \frac{\text{m}^2}{\text{s}} \text{ (such that } S_z = 0.01), \quad \gamma_x = \frac{\nu_z \Delta x^4}{(\pi \Delta z)^2} = 1.13 \times 10^8 \frac{\text{m}^4}{\text{s}}$$

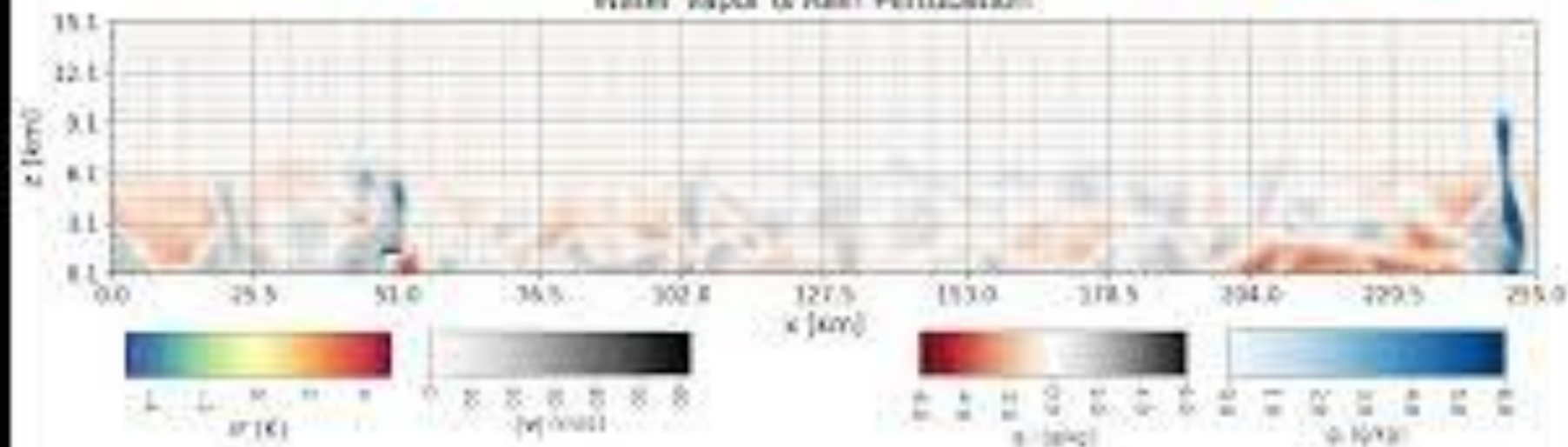
2D Scattered Convection

Solution at $t = 12.19$ hrs

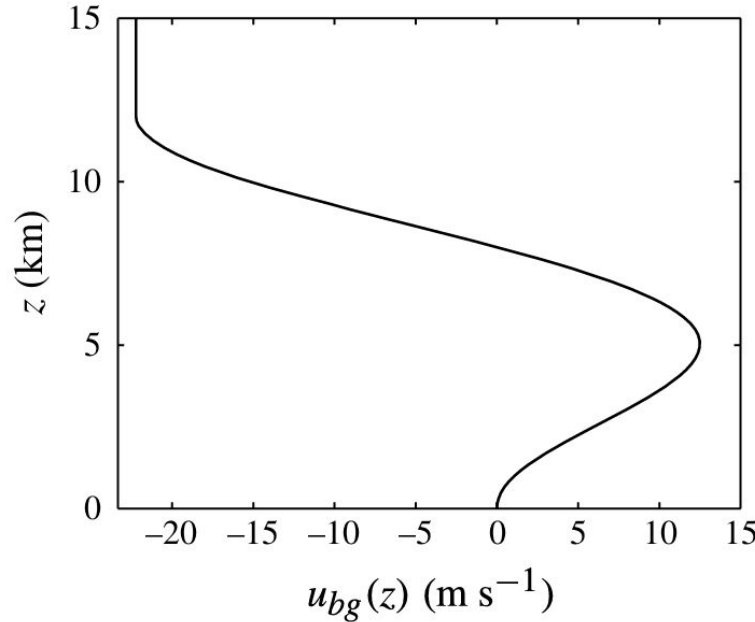
Temperature & Velocity Perturbation



Water Vapor & Rain Perturbation



Squall Line: Initial/Forcing Conditions

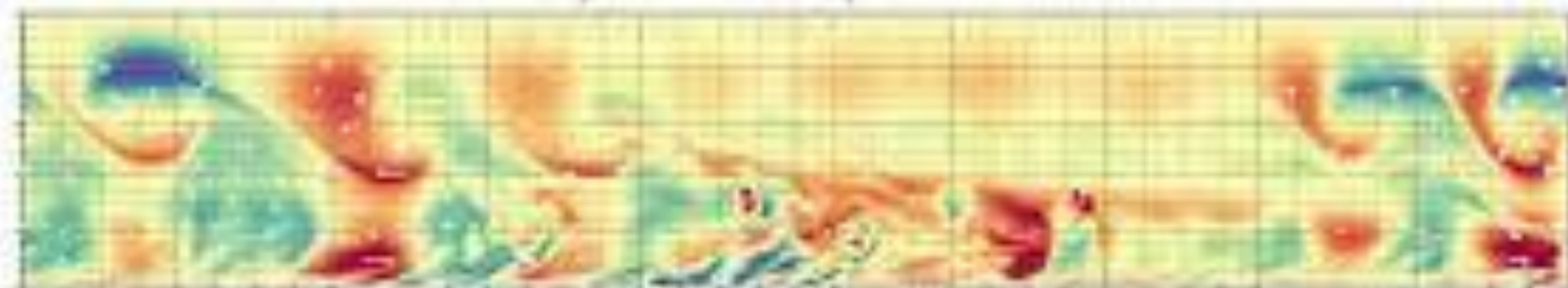


$$u^0(z) = u_{bg}(z) = \begin{cases} a \left(\cos\left(\frac{\pi z}{H_0}\right) - \cos\left(\frac{2\pi z}{H_0}\right) \right) & \text{if } z < H_0 \\ -2a & \text{if } z \geq H_0 \end{cases}$$

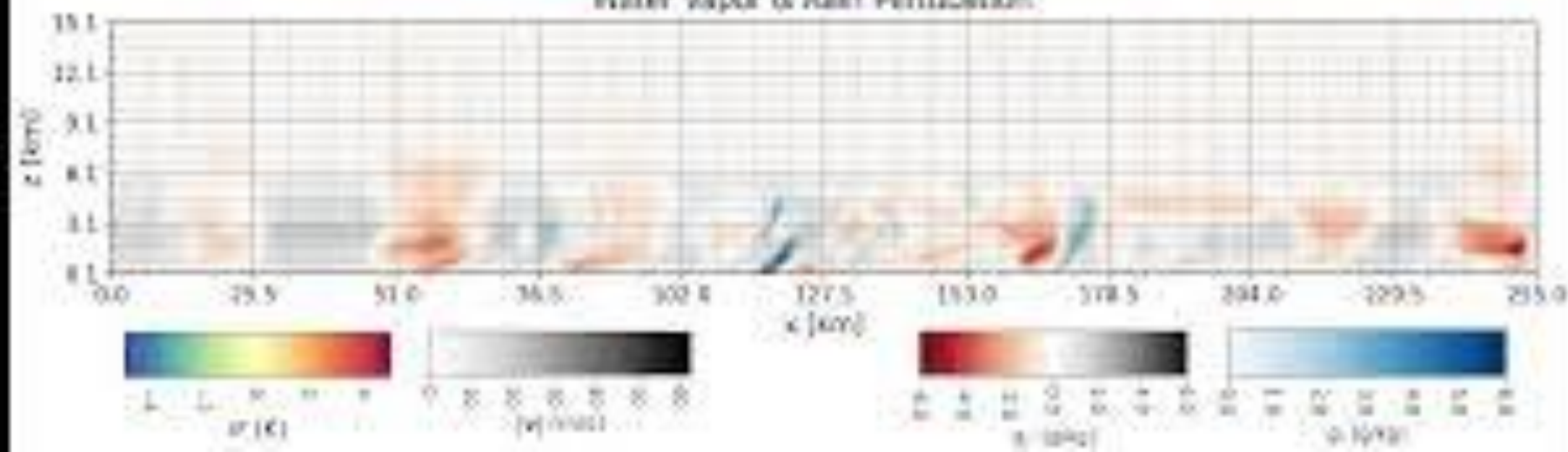
2D Squall Line

Solution at $t = 10.49$ hrs

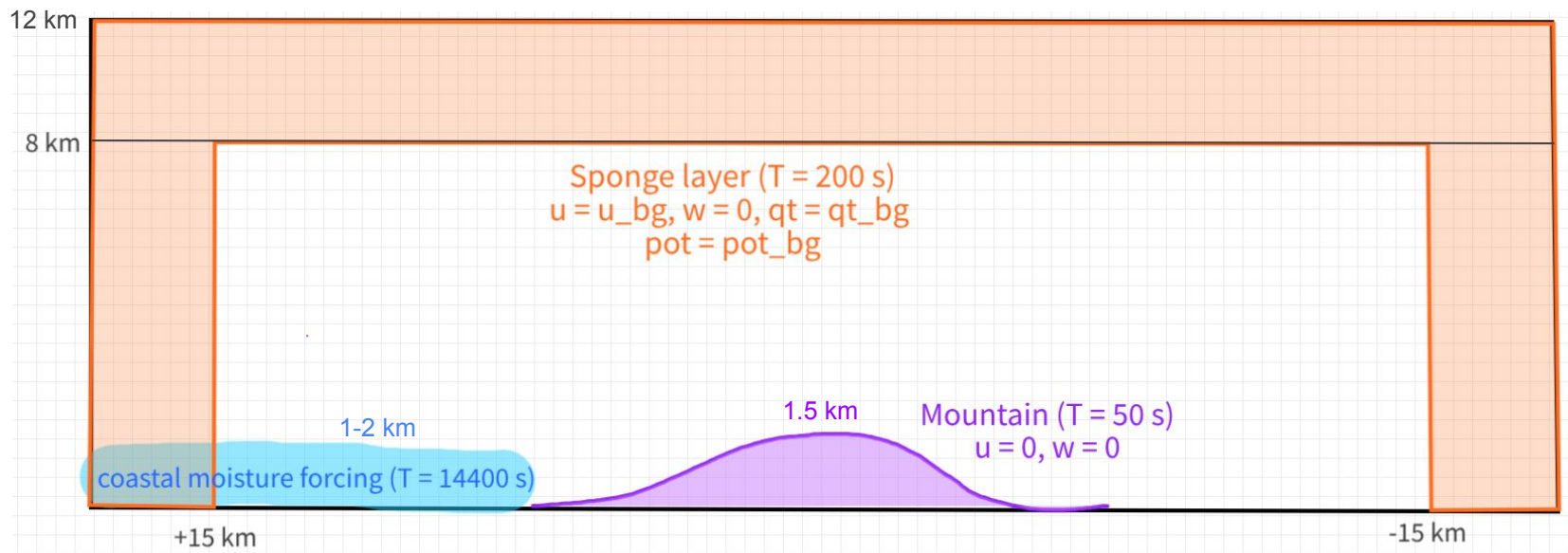
Temperature & Velocity Perturbation



Water Vapor & Rain Perturbation



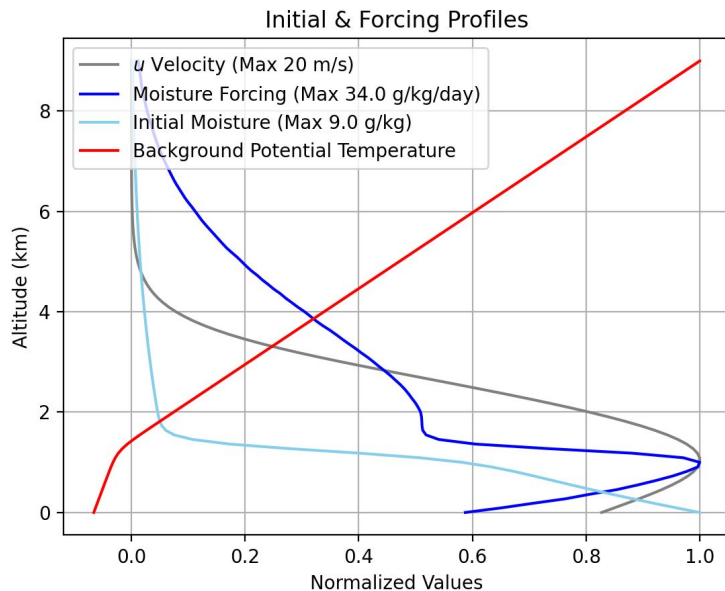
Atmospheric River: Sponge Mountain



$$\frac{\theta^{n+1} - \theta^*}{\Delta t} = \frac{M}{\tau_M} (\theta^{n+1} - \theta_M) \quad \longrightarrow$$

$$\theta^{n+1} = \frac{\Delta t}{\tau_M} \left(\frac{\theta^* + \theta_M M}{1 + M} \right)$$

Atmospheric River: Initial/Forcing Conditions



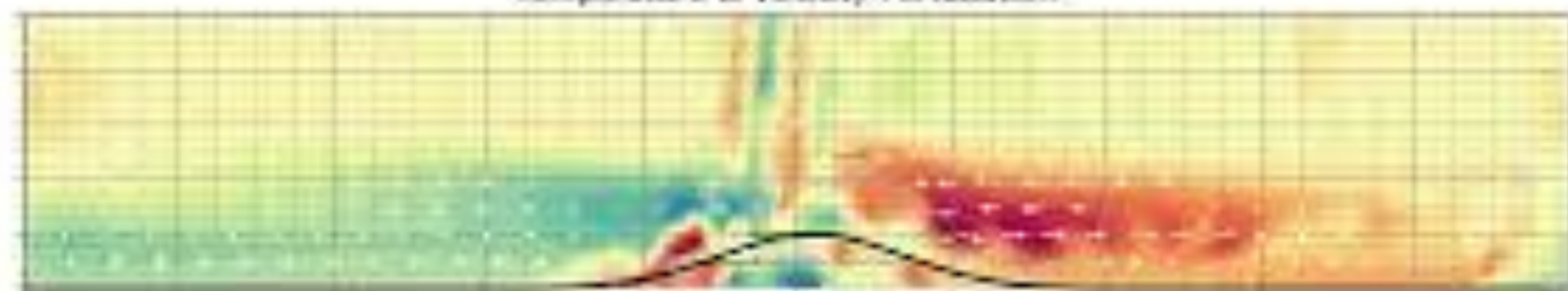
In addition to the above, we set the bottom u BC to free-slip (Neumann 0).

*Used *Gemini* to help set up representative background conditions.

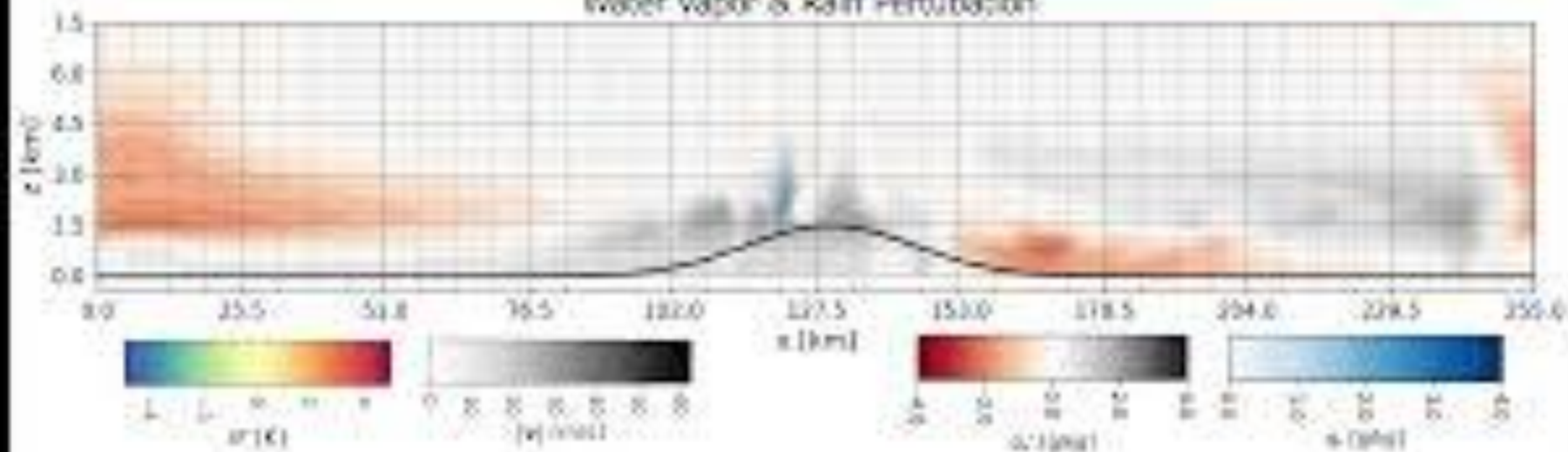
2D Atmospheric River

Solution at $t = 9.67$ hrs

Temperature & Velocity Perturbation



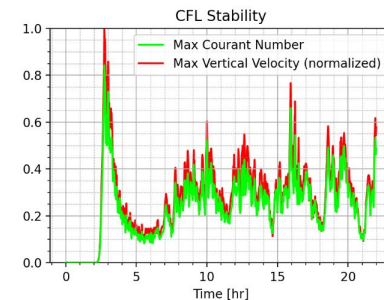
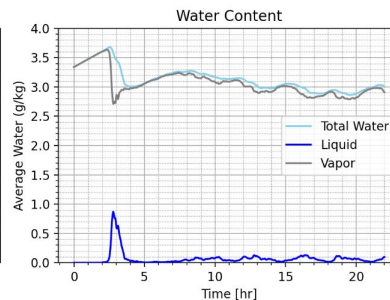
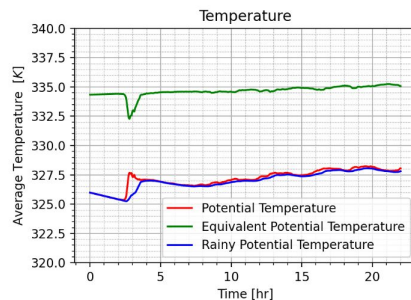
Water Vapor & Rain Perturbation



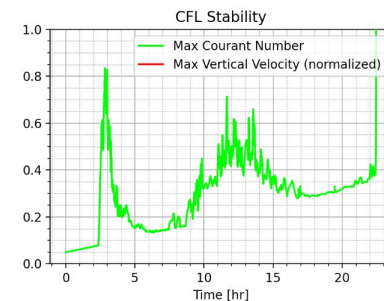
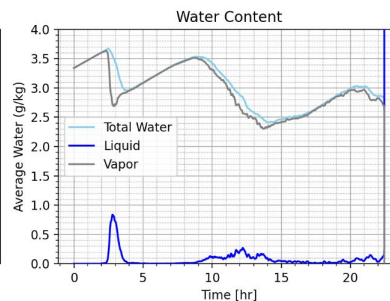
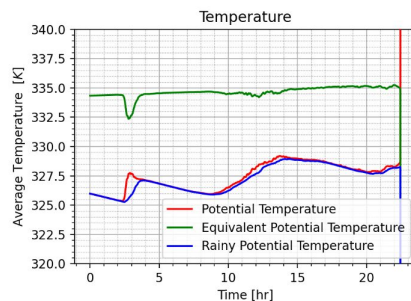
Conservations

- θ_e is conserved well due to there being relatively little rain compared to water vapor.
- CFL stability makes or breaks the sim (it later broke the squall run)
- KE and PE (not shown) aren't well-conserved due to external forcing

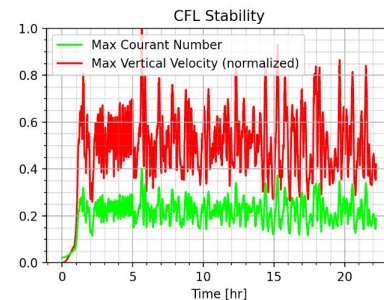
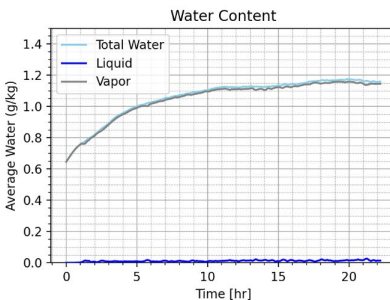
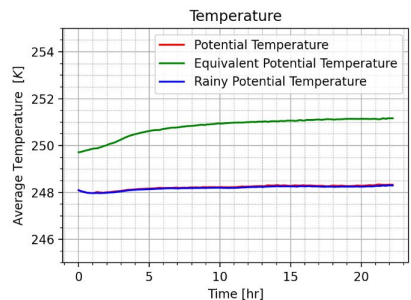
Scattered Convection



Squall Line



Atmospheric River



**Thanks for
listening!**